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The semiconductor premium: EM equity and the concentration risk within

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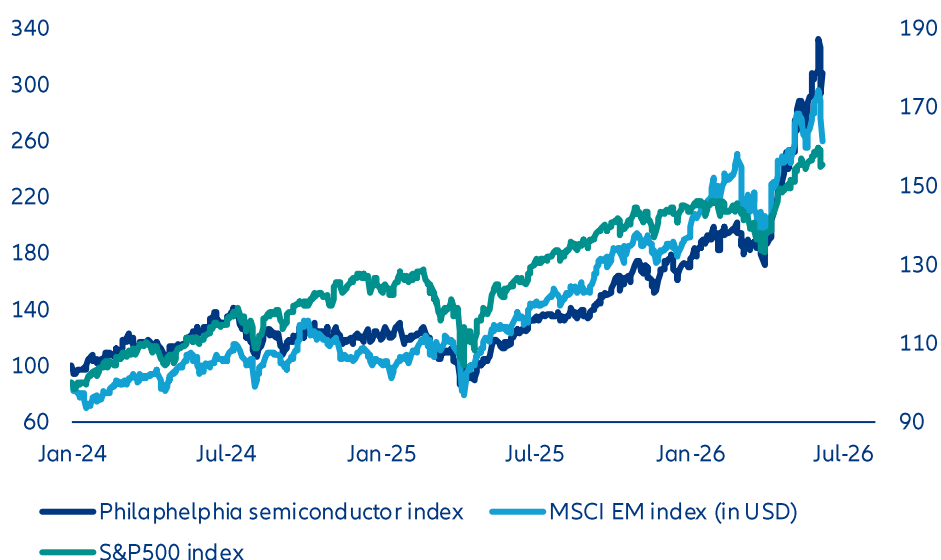
In Summary

- **Semiconductors are the backbone of the 2026 AI rally.** Capital investment is overwhelmingly directed at AI infrastructure, and the data centers doing the heavy lifting run entirely on chips. The numbers speak for themselves: the Philadelphia semiconductor index is up +90% YTD – twelve times the S&P 500's +7.5% - despite a couple of recent sell-off episodes in June. In emerging markets, which concentrate most of the supply chain, markets like South Korea and Taiwan are up +110% and +60% this year.
- **Concentration is the flip side of the boom but fundamentals are strong.** Concentration is most acute in EMs, where dependence on a single industry is now extreme. If the chips sector disappoints, the re-rating could be brutal. But the earnings are real and lopsidedly so: chips deliver 13% of S&P 500 profits on just 5% of sales, half of South Korea's profits on a fifth of its sales and over 70% of Taiwan's earnings and revenue alike. Ferocious demand colliding with supply bottlenecks from Middle East tensions are keeping prices sky-high.
- **The bull case still stands on five pillars.** First, hyperscalers show no sign of easing off. Upward capex guidance in Q1 points to a still-aggressive stance that should lift Asian chipmakers' profits by a remarkable +70% CAGR through 2026-2029. Second, the demand pipeline looks structurally solid: Broader AI diffusion, the corporate shift to agentic platforms and the memory-hungry new chip generation all reinforce one another. Third, the moat is real: capital and expertise are needed to build a modern foundry, keep competition at bay and pricing power intact. Fourth, valuations are compelling as based on solid earnings trajectory, and recent sell-offs offer new entry point on top picks. And fifth, in the over-concentrated EM space, semiconductors ensure geography and supply-chain position diversification, with an attractive trade-off compared to developed markets.
- **But strong convictions do not mean complacency: Four risks could unsettle the rally over the next 12-24 months.** First, a Middle East re-escalation could disrupt energy routed to Asia, forcing production halts – temporary or not. Second, bottlenecks in the Asian supply chain for inputs critical to data-center fleets could lift costs and lengthen lead times, throttling AI capex momentum. Third, a timing mismatch looms: New supply could come online just as capital allocation decelerates, should a fresh rate cycle take hold in the US or Europe. Fourth, competition is sharpening fast – Beijing is piling on funds and pressure to drive chip self-sufficiency (70% target in AI chips in 2026) and a bigger global footprint for its champions. Contrary to what market volatility suggests, we do not see lower inference cost to be a negative driver for memory chip business, but rather a guarantee of a sustainable book order if it comes with stronger AI diffusion and usage.

Semiconductors are driving the 2026 AI equity rally – and concentrating its risk

Equity markets are roaring back in 2026, with semiconductors as the primary engine. A brutal tech sell-off erupted last autumn, fueled by a rotation out of hyperscalers as the "AI bubble" narrative gained traction. The pain extended well into this year as frontier AI models – armed with native Customer Relationship Management (CRM), cybersecurity and data-processing capabilities – triggered a sharp de-rating of traditional software names suddenly fighting for relevance. Yet through the storm, one segment barely flinched: semiconductors. Quarter after quarter, chip makers delivered earnings that kept beating expectations, rewarding investors who held their nerve with relentless outperformance. The Philadelphia Semiconductor Index tells the story plainly, up 90% ytd, dwarfing the S&P 500's modest +7.5% gain and cementing semis as the single biggest beneficiary of the AI equity rally. But the most eye-catching trade of this cycle is unfolding not in the US but across emerging markets: The MSCI EM index has surged +23.2% YTD, and the engine behind that re-rating is, again, semiconductors as Asian chip and advanced packaging names ride the same AI capex supercycle. The investment wave is global and equity markets are pricing it that way.

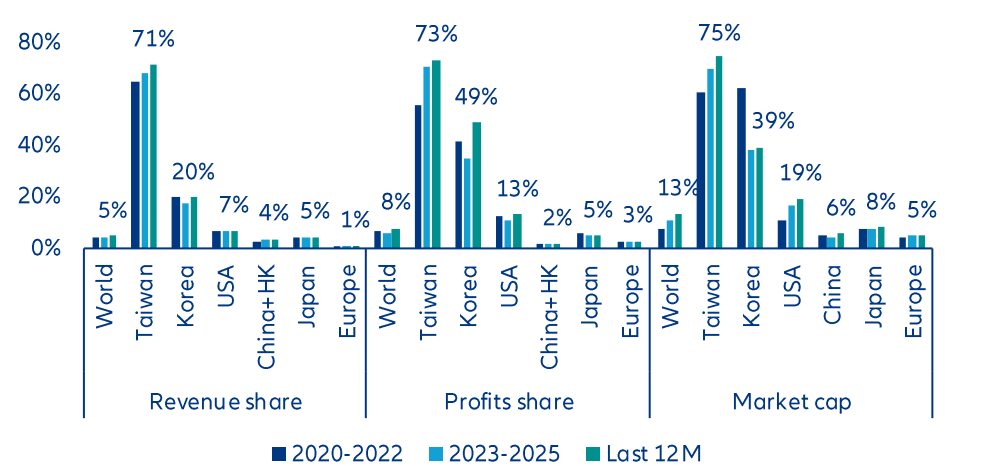
Figure 1: Equity index performance, rebased to 100 on January 2024



Sources: LSEG Datastream, Allianz Research

Semiconductors have quietly become the backbone of global equity markets – and a structural vulnerability at their core. Taiwan and South Korea stand in a league of their own. Between them, they concentrate the world's most critical foundry capacity and their stock markets reflect it starkly – Taiwan's semiconductor sector commands a 75% market cap share of its equity universe, South Korea's a still-commanding 39%, dwarfing every other major index. Yet the paradox lies in the revenue line: Semiconductor revenue shares across most exchanges remain surprisingly moderate, exposing an industry structured around a handful of dominant Asian foundries towering over a fragmented ecosystem of smaller players. Europe and the US have built their digitalization ambitions – and increasingly their AI infrastructure – on an Asian supply chain they do not control, a dependency that geopolitical stress-testing has already proven fragile.

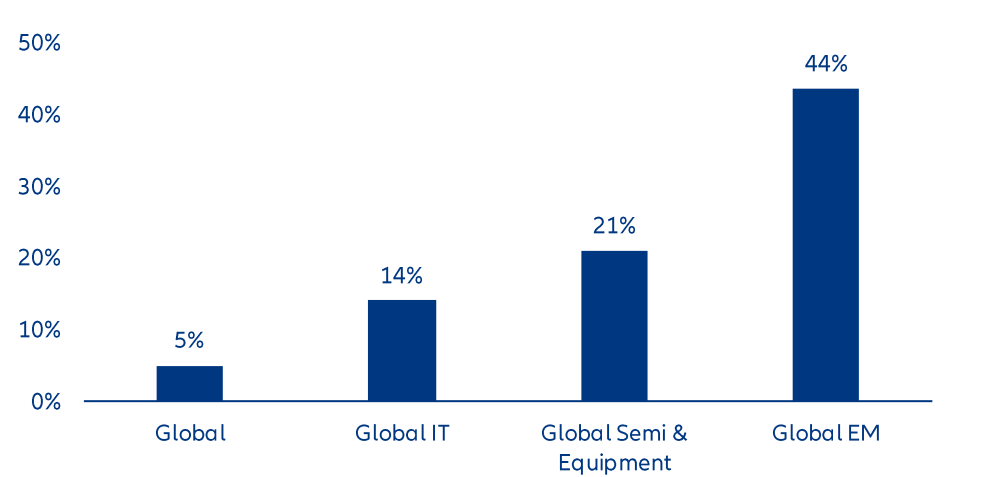
Figure 2: Semiconductor industry performance and footprint in the EM equity universe



Sources: LSEG Datastream, Allianz Research

Concentration is extreme – and the risk is double-layered. Taiwan's semiconductor profit share within its equity universe has vaulted from 55% in 2020-2022 to nearly 75% over the last 12 months. South Korea has mirrored that trajectory, climbing from 40% to close to 50% over the same period. This is not a gradual drift; it is a structural repricing of an entire industry's earnings power, driven by AI-fueled demand and ferocious pricing discipline from a handful of dominant foundries. But that concentration cuts both ways, and for EM investors, the vulnerability is not single-sided. MSCI EM already carries a 37% sector weight in IT and a 45% combined geographic exposure to Taiwan and South Korea alone, meaning the index is structurally pre-loaded with semiconductor risk before a single stock is picked. Stack on top of that the surging profit dominance of semis within those very markets and MSCI EM has effectively become a leveraged proxy on the AI capex cycle – a winning position today but a precarious one if sentiment toward the AI theme softens, earnings surprise cycles turn or geopolitical disruption bites into Asian supply chains. Any de-rating of the semiconductor complex would hit EMs through two channels simultaneously – sectoral and geographic – amplifying drawdown risk well beyond what headline index diversification would suggest. The US tells a more nuanced story: Semiconductor profit share is also at a cycle peak – approaching 20% – yet its broader and deeper equity market provides a natural buffer, diluting the concentration risk that EM investors simply do not have access to. The same industry driving the most exciting equity returns of this cycle is simultaneously the single biggest source of unpriced fragility within it.

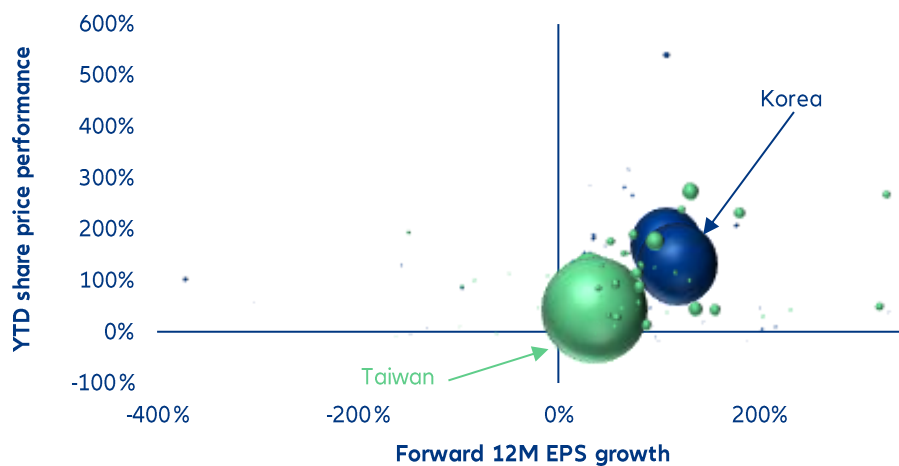
Figure 3: Aggregated market share of South Korea and Taiwan in MSCI index (as of April 2026)



Sources: MSCI, Allianz Research

Hyperconcentration of profits does not mean an absence of opportunity. South Korea dominates High Bandwidth Memory (HBM), the memory architecture at the heart of every AI accelerator, stacking dies vertically to deliver the massive low-latency bandwidth that large language models cannot function without. Taiwan holds what is effectively an unchallengeable near-term monopoly on advanced logic at 2-3nm node, where compute performance, miniaturization and energy efficiency converge – a process gap that Intel and SMIC are years from closing. The positive externalities from these twin centers of excellence ripple across equipment makers, packaging specialists and testing firms, generating an investment universe that extends well beyond the obvious names. That breadth is visible in the numbers: Median forward earnings growth recorded mid-May stood at 52% in South Korea and 42% in Taiwan, up sharply from 35% and 26% at the start of the year, a broad-based acceleration confirming that AI demand is diffusing through the entire supply chain, not just concentrating at its peak.

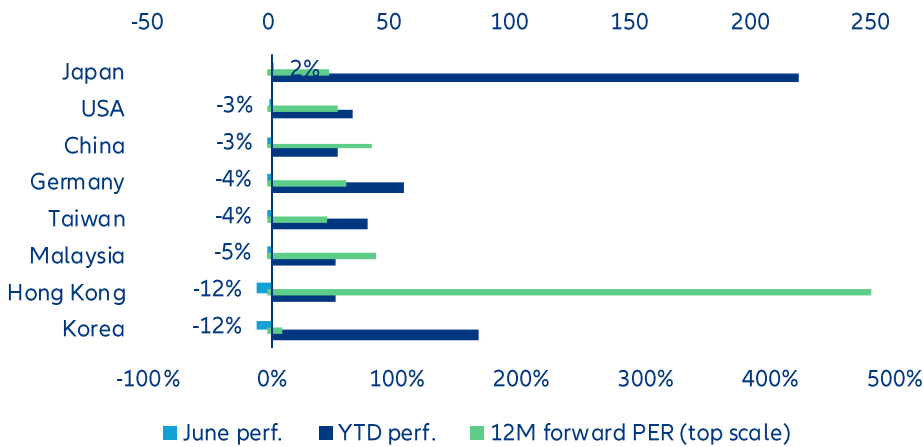
Figure 4: Forward earnings growth vs. 2026 share performance from South Korean and Taiwanese semiconductor firms



*Bubble size refers to market capitalization weight within national semiconductor universe.
Sources: LSEG Datastream, Allianz Research*

Volatility increases when you are under spotlights: go beyond day-to-day analysis. The -12% correction in June in global semiconductor equities was a positioning shock, not a fundamental one. It was caused by a sequence of negative signals that offer a reason for investors to cut their positions and take back their profits after an outstanding run (+82% YTD even after the sell-off). Among main downside catalysts, the disappointment caused by the non-revision of earnings trajectory from a top US chipmakers coupled with rising odds of a rate hike in USA and new concerns toward a weaker demand for HBM chips caused by the release of new Anthropic's Claude 4.8 opus that displays a more efficient inference cost model participate to pinch investor's sentiment toward semiconductor stocks. In reality, the damage was narrow rather than broad-based, concentrated in the two highest-beta markets — Korea, where the KOSPI fell 8.3% and tripped a circuit breaker, and Taiwan, where the largest foundry worldwide posted its largest-ever intraday drop. Korea bore the brunt for structural reasons: index hyper concentration in Samsung and SK hynix, heavy leverage, foreign outflows and a won past 1,560/USD, a mix that turns global tech wobbles into forced-deleveraging cascades. Yet the rally's anchors were not impaired — supply-chain demand kept accelerating and the megacaps rebounded fast — and valuations stay undemanding, with Korea near 10–11x forward earnings, the cheapest of the major Asian markets and North Asia broadly at a discount to the US.

Figure 5: Performance of semiconductor market and forward valuation per region breakdown

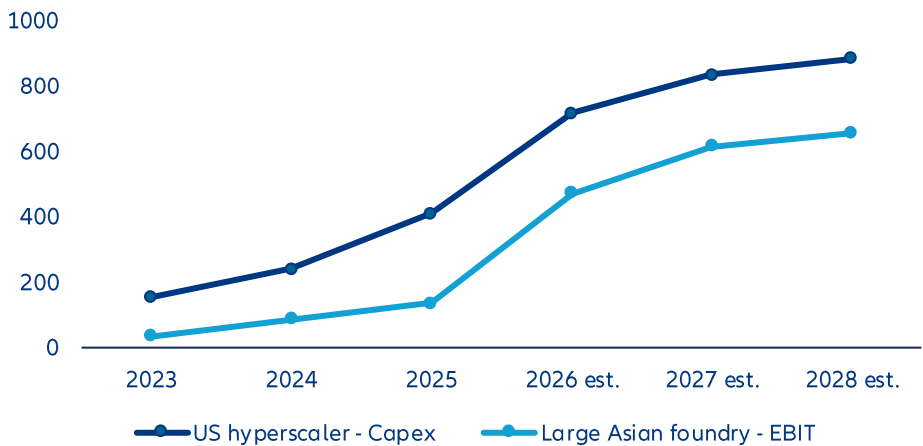


Sources: LSEG Datastream, Allianz Research

Semiconductors remain the purest expression of the AI capex cycle

The current phase of AI development is overwhelmingly infrastructural. Nations, hyperscalers and corporates alike are locked in a race to accumulate the largest possible compute capacity, with digital sovereignty increasingly measured by the volume of cloud space owned and the raw processing power available to run frontier models. Every data center announcement is, in essence, a chip order. US hyperscaler capex has surged from USD150bn in 2023 to over USD400bn in 2025, with guidance pointing to nearly USD900bn by 2028 – a near-sixfold expansion in five years. Large Asian foundry EBIT has tracked that trajectory with striking fidelity, albeit with a one-to-two-year lag that reflects the natural conversion time from capex commitment to wafer demand to earnings delivery: From a negligible USD35bn in 2023, foundry profits are projected to reach USD650bn by 2028. The lag is a feature, not a friction: It means the earnings wave is still building. With no credible sign of hyperscaler capex fatigue, the foundry profit outlook remains extraordinary. The AI-powered world being built is, above all, a chip-powered world.

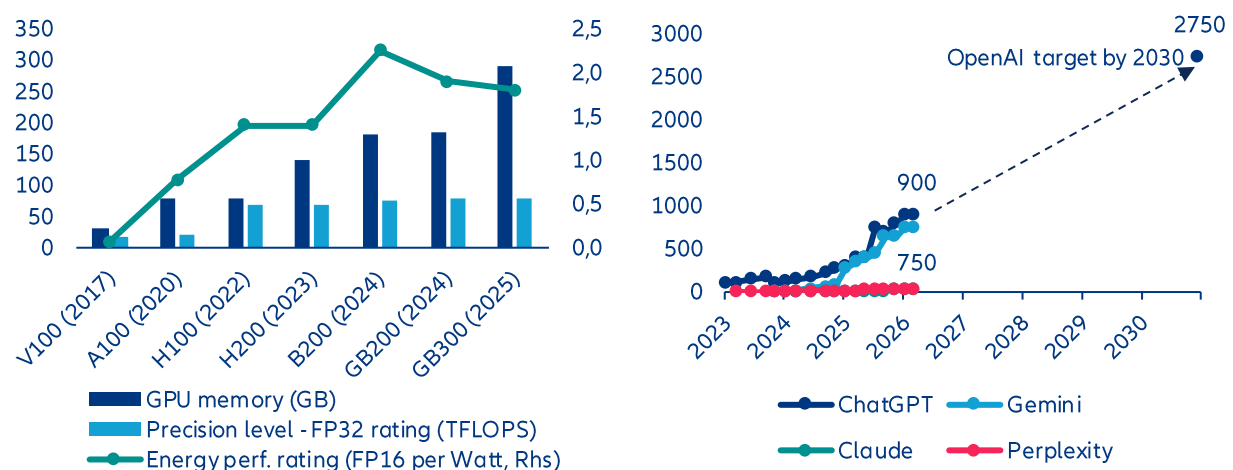
Figure 6: Guidance of US hyperscaler capital expenditure and large Asian foundry EBIT



Sources: LSEG Datastream, Allianz Research

AI diffusion is deepening, not plateauing – and memory demand follows. Each successive Nvidia GPU generation has delivered a step-change in memory capacity: from 32GB on the V100 in 2017 to 280GB on the GB300 in 2025, while energy performance per watt has expanded nearly twentyfold over the same period – a relentless hardware escalation that has made every frontier model generation structurally more chip-hungry than the last. The shift in AI workloads from training toward inference – often cited as a potential headwind for memory demand – is a false concern at the system level: what inference loses in per-query intensity, it more than recovers through sheer volume explosion. OpenAI alone is targeting 2.75bn users by 2030, implying that nearly one in four people on the planet will be regularly querying a frontier model – a demand profile that makes aggregate memory chips Dynamic Random Access Memory (DRAM) and Not-And (NAND) consumption essentially non-discretionary infrastructure spending. Unlike software incumbents facing direct substitution from AI-native capabilities, semiconductor and memory suppliers sit upstream of every model, every query and every data-center expansion – insulated from the application-layer disruption playing out below them. The democratization of AI access is not a risk to the chip cycle; it is its most powerful demand multiplier. The efficiency-bear case that used to come back at our ears at each release of new frontier AI model assumes demand is fixed, but it rarely is: cheaper, leaner inference is precisely what pulls AI into more users, more always-on agentic and multimodal workloads, and more applications that were previously uneconomic — so falling cost per query tends to expand aggregate query volume faster than it compresses the memory intensity of each one. Crucially, the memory call is driven by the scale of deployed infrastructure rather than the elegance of any single model: every incremental accelerator shipped is loaded with HBM and DRAM, and the shift toward inference and long-context, agentic workloads actually raises KV-cache memory requirements, meaning a broader, more efficient installed base de facto consumes more chips in aggregate, not fewer.

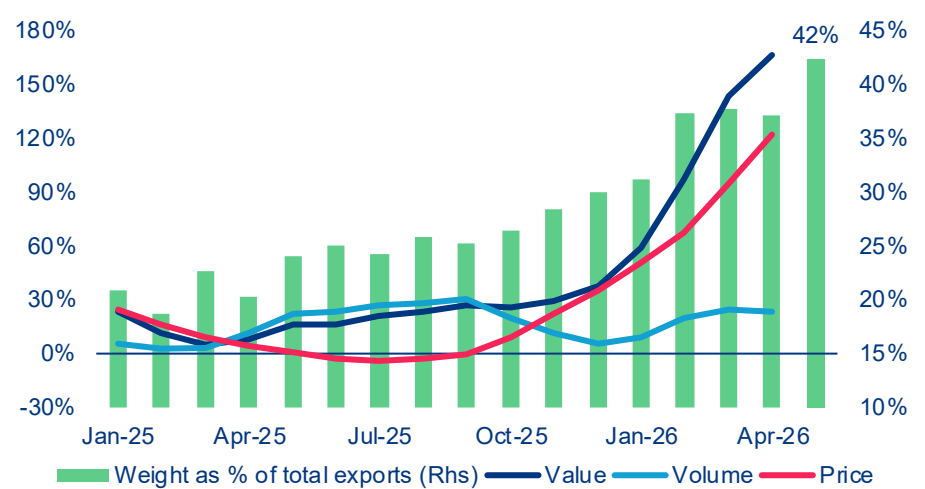
Figure 7 : Memory capacity, performance and power efficiency of GPU (based on Nvidia models) / Number of users of main LLM models, current and estimated (in mn of users)



Sources: Nvidia corporate brochure, Allianz Research

Supply tightness is extending pricing power – and acting as a macro hedge. Supply was already structurally tight before geopolitical stress entered the equation. The Middle East conflict has added an acute dimension: disruption to helium supply chains – a critical purification gas sourced partly from Qatar and indispensable to wafer fabrication – triggered precautionary front-loading across South Korean and Taiwanese buyers, compressing available inventory precisely when AI-driven demand is at its most insatiable. The result is visible in spot prices with brutal clarity: Flash NAND 64GB spot prices have nearly tripled in 2026 alone, surging from below USD5 to nearly USD20 per unit – a pass-through dynamic that semiconductor firms have executed without friction, reflecting a pricing power that few industries can match. When the end-demand justification is AI productivity gains worth trillions in enterprise value, customers absorb chip price inflation with remarkably little resistance. South Korea's semiconductor export value growth has accordingly exploded to +150% year-on-year – a magnitude that dwarfs volume growth and confirms that this is a margin and pricing story as much as a volume one, with the spread between value and volume growth at its widest on record. The slow progress in new capacity build-up postpone in time breakeven point between supply and demand, leading to a likely extension of memory shortage until 2027. As a result, the mix of supply imbalances coupled with extremely-high concentration of the segment ensure a strong pricing power Asian chipmakers that should help them keeping a high margin profile for long.

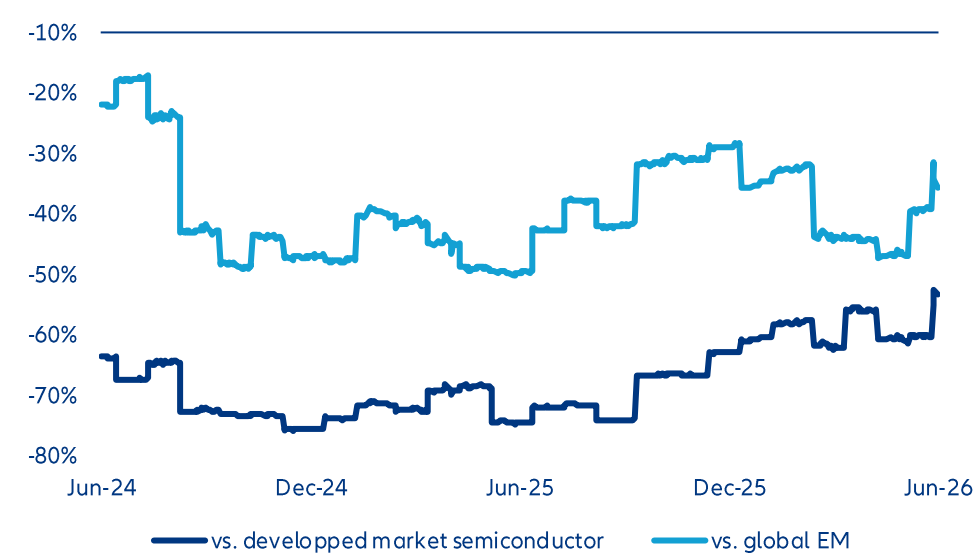
Figure 8: South Korea semiconductor exports trend (price, value and volume) and weight



Sources: Bank of Korea, LSEG Datastream, Allianz Research

Valuations remain compelling despite rally. Even after the year-to-date rally, EM semiconductors remain conspicuously cheap on a market-weighted basis: as of June-26 the industry still trades at roughly a 55% forward-P/E discount to developed market (DM) semiconductor peers and around a third below broad global emerging market benchmark (EM). The rally has admittedly started to dilute that gap — both discounts have compressed off their late-2024 troughs (circa -75% versus global semis, -50% versus global EM), with the narrowing accelerating sharply in the most recent leg, particularly against global EM. Yet the residual discount to DM names sits oddly with the reality that EM players now occupy the most critical, highest-value nodes of the chain — leading-edge foundry and HBM — a positioning that arguably warrants convergence rather than a persistent markdown. Part of the optical cheapness is also a concentration artefact: the aggregate multiple is dragged down by a handful of mega-caps whose steep forward-earnings trajectories compress the denominator, so the headline discount reflects those few names rather than uniform value across the sector. Beyond the pure valuation call, that same skew opens an arbitrage-and-diversification angle — the better risk-reward may sit further along the supply chain in equipment, materials, advanced packaging and testing, rather than in the crowded foundry and pure-memory leaders.

Figure 9: Valuation premium/discount of EM semiconductor industry vs. global EM and semiconductor industry (based on forward 12M price to earnings ratio)



Sources: LSEG Datastream, Allianz Research

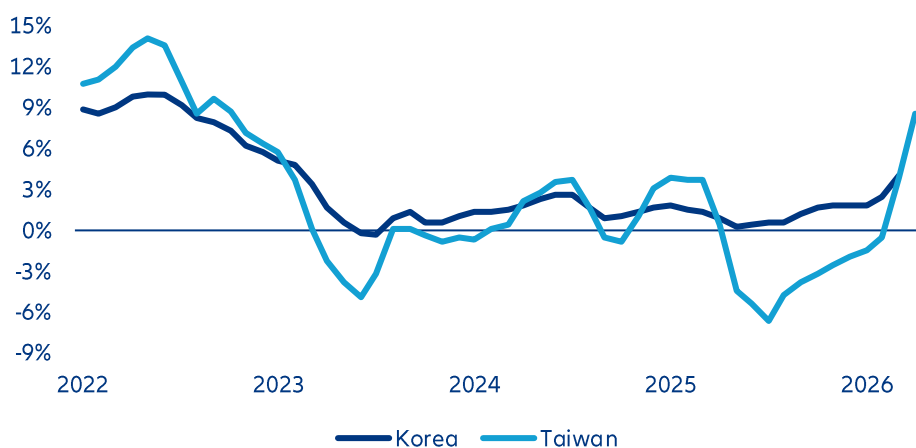
Four risks that warrant active monitoring

Despite the compelling semiconductor narrative, complacency would be a mistake. A clear-eyed view of the next 12-18 months reveals four factors that deserve active monitoring before the next leg of the rally is taken for granted.

Risk 1 - Middle East disruptions: when scarcity becomes a double-edged sword

The same supply tightness that has gifted semiconductor firms extraordinary pricing power now carries a sting in the tail. Helium storage levels have dropped to uncomfortable lows as tensions persist. Compounding the picture, wafer component costs and freight rates have moved meaningfully higher, inflating production costs at a moment when delivery pipelines are already stretched. The structural constraint is unforgiving: New fabrication lines outside Asia take years to reach operational relevance, leaving current capacity with little slack to absorb sustained disruption. Q1 earnings provided some relief – margin guidance pointed only to modest downward adjustments rather than any structural deterioration – and the industry is not defenseless: three to six months of secured helium inventory, combined with recycling protocols put in place as far back as 2022, provide a meaningful buffer against near-term shocks. But that buffer has a clock on it. Should Middle East tensions persist and logistical disruption deepen, the scarcity narrative that has been a profit tailwind could rapidly invert into a production bottleneck – with spot price spikes masking volume shortfalls that ultimately disappoint revenue expectations. The PPI chart captures the inflection already: Taiwan's production cost index has snapped from -7% year-on-year in late 2025 to nearly +9% by mid-2026 – the sharpest reversal in the dataset – while South Korea's is also accelerating, signaling that cost pressures are landing on balance sheets now, not as a future risk. For an industry priced for perfection, that reversal is worth watching with precision.

Figure 10: Production cost index in South Korea and Taiwan (PPI, annual change in %)



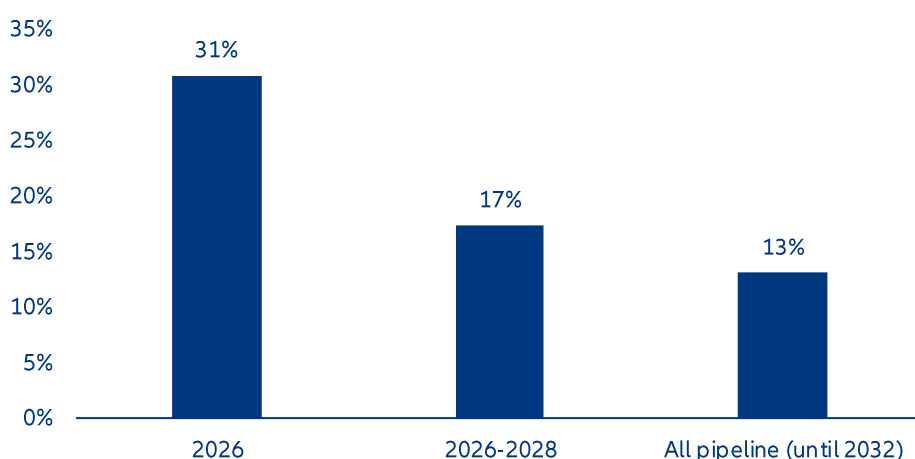
Sources: LSEG Datastream, Allianz Research

Risk 2 - AI capex execution gap: when the infrastructure pipeline meets real-world friction

The AI capex supercycle is the single most powerful demand driver underpinning the semiconductor bull case – which makes any credible threat to its execution timeline a risk that cannot be dismissed. As of end-January 2026, only 31% of US data-center capacity committed for delivery in 2026 was officially under construction – a figure that drops to 17% for the 2026-2028 window and a sobering 13% across the full pipeline to 2032. Behind the relentless drumbeat of hyperscaler capex announcements lies an execution gap that is widening, not closing, driven by planning bottlenecks, grid connection delays, construction labor shortages and permitting friction that no balance sheet commitment can shortcut. A data center that is announced but not built does not pull forward a chip order.

The risk compounds if the macroeconomic backdrop shifts. A renewed inflationary impulse – entirely plausible given rising freight costs, energy price pressure from Middle East tensions and sticky services inflation – could force another monetary tightening cycle, pushing long rates higher at precisely the moment when capital-intensive infrastructure projects are most rate-sensitive. Higher rates do not just raise discount rates on future cash flows, they make banks materially more restrictive in extending project finance to private data-center developers, the fastest-growing segment of the pipeline. Given the near-perfect correlation between hyperscaler capex trajectories and Asian foundry profit guidance, any sustained flattening of investment commitments would transmit rapidly into semiconductor earnings downgrades, eroding pricing power simultaneously as the demand urgency that has allowed cost pass-through begins to dissipate. For a sector priced at a meaningful premium, the margin for disappointment is narrow.

Figure 11: Percentage of the US data-center pipeline officially under construction



Sources: Sightline Climate, Allianz Research

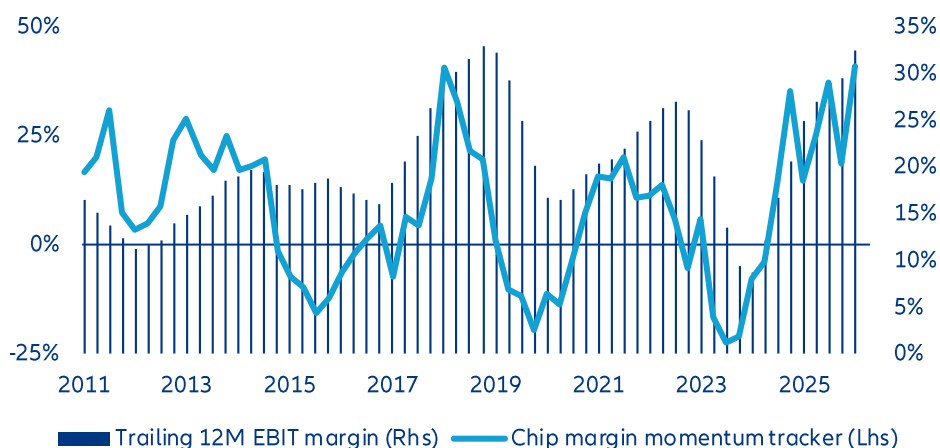
Risk 3 - The capex timing trap: will history rhyme?

The semiconductor industry has a well-documented and costly habit: it invests in new capacity at the top of the cycle, and the production lines come online just as demand rolls over. The pattern is visible with painful regularity in Figure 11: EBIT margins and chip margin momentum have oscillated in violent cycles, with the most brutal compression episodes – 2015, 2019, 2023 – coinciding precisely with waves of new supply hitting markets into weakening end-demand. The post-2022 episode is the most instructive: A surge of capacity investment triggered by pandemic-era shortage hysteria delivered production lines into a demand vacuum, crushing margins to near zero in 2023 and generating inventory write-downs across the industry. The question is whether 2027-2028 will rhyme. The timing risk is concrete and quantifiable. TSMC's Arizona Fab 21 Phase 2 – targeting 3nm and 2nm production – is scheduled for completion by 2027-2028, alongside its ongoing Japan facility expansion in Kumamoto targeting mature nodes. Samsung's P3 and P4 lines in Pyeongtaek – among the world's largest semiconductor complexes – are expected to reach full operational capacity over the same window, adding substantial HBM and advanced logic output to global supply. SK Hynix's M15X facility in Cheongju, purpose-built for next-generation HBM production, is similarly slated for ramp-up through 2027. Collectively, these projects represent the largest synchronized capacity expansion in the industry's history, hitting markets simultaneously, with limited ability to delay or redirect output once fab construction reaches completion.

Trailing 12-month EBIT margins are approaching 30-32%, levels last seen fleetingly around 2018 and never sustained beyond a few quarters. Chip margin momentum, the leading indicator, is already showing the first signs of oscillation at elevated levels, a pattern that historically has preceded compression. The bull case argument – that unprecedented data-center pipeline demand will absorb the incoming supply wave – is plausible but carries the execution uncertainty documented in Risk #2: only 13% of the US data-center pipeline to 2032 is currently under construction. If the demand realization timeline slips by even 12-18 months relative to supply delivery, the industry faces the same inventory build and margin compression dynamic it has experienced in every prior cycle.

Semiconductor profits have been the fuel of the EM equity rally; a cyclical mean-reversion in foundry margins would not just slow the trade, it would reverse it.

Figure 12: EBIT margin and margin momentum index of large Asian foundries



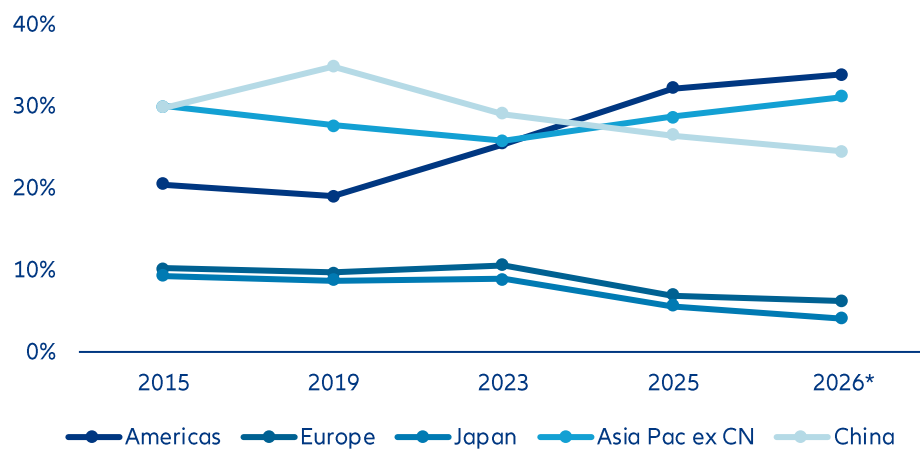
Sources: LSEG Datastream, Allianz Research

Risk 4 - The China wildcard: a competitor in construction, not yet in contention

Figure 13 below frames the competitive landscape with striking clarity: Americas' share of global semiconductor shipments by value has surged from 20% in 2015 to 34% in 2026, while Asia Pacific ex-China has climbed to 31% – the two dominant and widening poles of the industry. China, meanwhile, has seen its share compress from 30% to 24% over the same period, a trajectory that reflects both export controls tightening its access to advanced process technology and a domestic industry still years from self-sufficiency. Europe and Japan have faded to near-irrelevance at 4-6%, a structural marginalization that makes their dependence on Asian supply chains all the more acute. But the direction of Chinese policy ambition is unambiguous and dismissing it entirely would be a mistake. Beijing has set an explicit target of covering 70% of domestic semiconductor needs through indigenous production – a goal backed by the most aggressive state financing program in the industry's history. The Big Fund Phase III alone mobilizes an estimated USD30-50bn, part of a total USD70bn incentive envelope that dwarfs both the US CHIPS Act (USD53bn) and the EU Chips Act (EUR43bn) in raw commitment size. The political signal is equally unambiguous: During President Donald Trump's recent visit, Chinese officials made clear they had no interest in dismantling semiconductor trade barriers. The strategic priority is building domestic capacity, not negotiating market access. Semiconductors have become a theatre of technological sovereignty, not commercial competition.

Yet the gap between ambition and execution remains vast across the three most critical chokepoints. On EUV lithography – the indispensable tool for sub-7nm node production – China has made headline progress, with reports of a prototype developed under a project drawing on expertise from former ASML personnel. But scaling from prototype to mass production is an entirely different challenge: The official target of EUV mass production by 2028 is viewed skeptically by industry experts, with more credible timelines pointing to post-2030 at the earliest for volume output. On High Bandwidth Memory, the process know-how required to compete with SK Hynix and Samsung at HBM3/HBM4 levels demands both multi-billion dollar capex and years of yield optimization that state funding alone cannot shortcut. And on EDA software – the design automation tools that underpin every advanced chip architecture – building competitive capability requires deep, accumulated human expertise that neither subsidies nor industrial policy can manufacture quickly. China is a competitor in construction, not yet in contention. But the 2028-2030 window, when domestic capacity begins reaching meaningful scale, is precisely when Asian foundry supply is also expected to surge. The collision of those two dynamics is a risk the market has not yet begun to price.

Figure 13: Global semiconductor shipments by value - geographic market share



Sources: SEMI, Allianz Research

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